$$\text{Overall complexity} = \underbrace{O(|\mathcal{E}|) + O(|\mathcal{V}_s| + |\mathcal{E}_{\mathcal{V}_s}|)}_{\text{Steps 1–2}} + \underbrace{\sum_S O(M|S|) + \sum_{S \text{ w/o BC}} O(M|S|D(S)) + \sum_{S \text{ with BC}} O(M|S|^2) + \sum_{S \text{ with BC}} O(2^{|S|}|S|M)}_{\text{sum over all SCC}} = O(|\mathcal{E}| + M|\mathcal{V}_s|D) + \textstyle\sum_{S \text{ with BC}} O(2^{|S|}|S|M)$$